

MOOSONEE, ON, Canada

WMO: 713980

Lat: 51.291N

Lon: 80.617W

Elev: 30

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -29.3	(c) -24.6	(d) -36.7	(e) 0.7	(f) -29.0	(g) -32.2	(h) 0.9	(i) -24.1	(j) 21.9	(k) 9.9	(l) 20.3	(m) 7.2	(n) 4.9	(o) 250	(p) 0.712

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 21.5	(c) 84.5	(d) 68.1	(e) 80.6	(f) 66.6	(g) 76.7	(h) 65.0	(i) 71.2	(j) 79.9	(k) 68.7	(l) 76.8	(m) 66.6	(n) 74.4	(o) 11.0	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 68.1	(b) 103.5	(c) 75.6	(d) 65.8	(e) 95.3	(f) 73.4	(g) 63.5	(h) 87.6	(i) 71.2	(j) 35.2	(k) 80.4	(l) 33.0	(m) 76.7	(n) 31.3	(o) 74.5	(p) 79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.0	18.5	16.4	-35.7	91.4	5.1	2.9	-39.3	93.4	-42.3	95.1	-45.1	96.8	-48.8	98.9
			-35.9	74.7	5.0	2.4	-39.5	76.4	-42.4	77.9	-45.3	79.2	-48.9	81.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	32.6	-1.3	1.0	12.9	28.4	43.7	55.0	61.4	60.2	52.6	40.8	26.1	8.5		
	DBStd	24.66	13.89	12.57	14.44	11.12	10.35	9.83	7.51	7.41	8.51	8.13	11.14	14.10		
	HDD50	7470	1590	1373	1152	652	259	55	2	5	67	310	718	1287		
	HDD65	11969	2055	1793	1617	1099	665	329	166	188	385	752	1168	1752		
	CDD50	1116	0	0	0	2	64	206	355	321	144	23	1	0		
	CDD65	138	0	0	0	0	5	29	53	38	12	1	0	0		
	CDH74	1410	0	0	0	1	97	377	500	338	93	4	0	0		
	CDH80	384	0	0	0	0	26	125	141	77	15	0	0	0		
	WSAvg	7.8	7.6	7.8	8.1	8.2	8.2	7.8	7.3	7.3	7.7	8.1	8.3	7.8		
	PrecAvg	28.5	1.5	1.1	1.3	1.7	2.2	3.0	3.9	3.2	3.3	3.0	2.6	1.6		
	PrecMax	43.7	3.0	2.4	3.9	3.8	4.1	7.0	7.6	7.1	6.9	6.8	5.7	4.3		
(15) Precipitation	PrecMin	21.3	0.4	0.2	0.3	0.4	0.5	0.6	0.7	1.2	1.5	1.1	0.9	0.4		
	PrecStd	5.8	0.7	0.6	0.8	0.8	1.0	1.5	1.7	1.4	1.2	1.2	1.1	0.7		
	0.4%	DB	36.5	39.1	53.0	66.3	83.7	89.8	89.2	86.2	81.8	72.5	56.2	39.7		
	MCWB	35.0	35.4	44.9	50.2	64.8	70.8	71.4	71.0	67.7	61.3	49.6	36.6			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2%	DB	32.3	33.0	44.8	55.9	75.2	83.6	83.8	82.1	76.5	64.3	47.9	35.9		
	MCWB	31.3	31.0	38.4	45.0	59.3	65.9	68.5	68.6	65.6	56.8	43.0	34.4			
	5%	DB	28.0	27.1	40.0	49.6	68.8	78.1	79.8	77.7	71.6	58.2	43.4	32.6		
	MCWB	27.0	25.5	35.6	41.8	55.9	64.0	66.9	66.5	63.0	53.1	40.5	31.5			
	10%	DB	21.2	21.8	35.1	44.1	62.4	72.6	75.3	73.5	66.9	52.8	39.5	29.1		
	MCWB	20.2	20.5	31.9	38.2	53.2	61.5	65.3	64.7	60.2	48.5	37.4	28.1			
	0.4%	WB	35.5	36.1	47.0	51.5	67.0	73.3	74.0	73.4	70.6	62.6	50.5	37.9		
	MCDB	36.2	38.3	50.3	61.5	78.7	84.3	84.9	81.5	79.1	68.5	54.4	39.2			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	31.7	31.1	39.4	46.2	61.0	69.2	71.0	70.5	67.0	58.0	44.0	34.5		
	MCDB	32.5	33.2	44.1	54.0	70.8	78.0	79.5	78.8	73.9	62.8	46.3	35.5			
	5%	WB	27.0	25.5	35.7	42.1	57.4	66.0	68.5	68.0	64.0	53.5	40.5	31.6		
	MCDB	27.9	27.0	39.3	49.1	66.9	75.2	76.3	75.8	70.8	58.0	43.4	32.6			
	10%	WB	20.2	20.5	32.0	38.5	53.6	62.8	66.3	65.5	60.8	48.9	37.6	28.1		
	MCDB	21.1	21.7	35.1	43.7	62.0	71.4	74.0	72.2	66.3	52.3	39.6	29.1			
	MDBR	19.0	21.8	22.8	19.4	20.8	23.3	21.5	19.9	18.3	13.5	12.1	15.7			
	5% DB	MCDBR	19.4	24.1	24.1	27.5	32.9	29.8	26.4	25.1	24.9	21.6	15.6	14.5		
(36) Mean Daily Temperature Range	MCWBR	18.9	22.3	18.7	18.2	18.8	16.4	14.4	14.2	15.7	15.2	12.6	14.0			
	5% WB	MCDBR	19.6	23.5	22.3	25.2	29.4	26.2	23.6	22.7	22.4	20.2	14.6	14.5		
	MCWBR	19.1	22.0	17.9	17.6	18.3	16.7	14.6	14.2	15.4	15.4	12.3	14.1			
	taub	0.260	0.273	0.288	0.351	0.370	0.376	0.399	0.382	0.366	0.310	0.299	0.257			
(40) Clear-Sky Solar Irradiance	taud	2.227	2.229	2.254	2.208	2.295	2.373	2.357	2.406	2.430	2.560	2.415	2.276			
	Ebn at Noon	240	272	290	280	280	279	270	269	260	260	227	224			
	Edh at Noon	22	29	34	41	39	36	37	33	29	21	18	17			
	RadAvg	390	761	1265	1623	1606	1833	1680	1376	977	574	301	261			
(44) All-Sky Solar Radiation	RadStd	43	52	91	159	136	142	118	101	74	63	32	38			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	+2.68	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page